The Arrow of the Reversible: Boltzmann's H and Loschmidt's Paradox

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The Removable Singularity Framework

Abstract

A 262,144-point ensemble is evolved by Arnold's cat map - a deterministic, exactly reversible dynamical system (det = 1, exact integer inverse on the lattice). (1) The measured Boltzmann H rises monotonically from ln(64) = 4.1589 to 8.3173 (saturation ln 4096) in 8 steps: a Second Law rising from a reversible law, no randomness introduced - Loschmidt's (1876) reversibility objection turned into its own evidence. (2) The same state seen through a 256x256 lens rises by the same clump-dilution ~4.16 nats: the measured arrow is a property of the coarsening, not the system (Gibbs 1902). (3) Eight inverse steps return every one of the 262,144 points to its exact starting site (0 deviations) and H to 4.158883 exactly: Zermelo's (1896) recurrence threat defused - the entropy that rose was never destroyed, only scattered across bin boundaries. The exact state has no entropy at all: with a lens fine enough, H = 0 for all time. The arrow of time is the 0/0 of measurement: reversible law + coarse lens = the Second Law; reversible law + fine lens = the identity. The lens spends hidden information as heat (the Ch.65 fee).

1. The Law Is Reversible (Liouville, exact on the lattice)

(x,y) -> (x+y, x+2y) mod L   det = 1

The cat map preserves phase-space volume exactly and has an exact integer inverse (2x-y, -x+y) mod L on the 2^18 x 2^18 integer torus. Forward then inverse = identity at every site - the same reversibility Noether demanded (Ch.66): the law cannot tell forward from backward.

2. The H-Curve (Boltzmann 1872)

H: 4.1589 -> 4.379 -> 4.709 -> 5.363 -> ... -> 8.3173 (+4.158 nats)

H(t) = ln N - (1/N) * sum n_i ln n_i over 64 x 64 coarse bins. Starting as a 512x512 clump (8x8 bins), the measured curve is monotonically non-decreasing, saturating at 8.3173 = ln(4096) minus a fluctuation tail. The rise of exactly 4.1584 nats equals ln 64: the information the clump concentrated is what the coarse lens sees dissipate. Deterministic, reversible dynamics produced an arrow - Loschmidt's protest (1876) dissolves into this fact.

3. The Lens (the coarse-graining 0/0, Gibbs 1902)

The identical state measured through a 256x256 nose saturates at 11.0393 against baseline ln(1024) = 6.9315: rise 4.108 nats, matching the 4.158-nat rise at 64x64. The arrow is the same clump-dilution in both noses; only the numbers change with the instrument. What the coarse lens spends as heat, the fine lens still sees as information in bin boundaries.

4. The Return (Zermelo 1896 defused)

inverse x8: 0/262144 deviations, H back to 4.158883 EXACTLY

Eight inverse steps restore every point to its starting lattice site: the histogram returns to uniform-in-64-bins and H to 4.158883 - exactly the start. The rise was a refraction, not a destruction: recurrence (Ch.57) and reversibility (Ch.66) hold inside the exact state, and the arrow is read only through the coarse eye. Loschmidt and Zermelo together lose to Gibbs' lens.

5. The 0/0 Proof

With a lens fine enough to isolate each lattice site, every point is alone in its bin: H = 0 for all time - the exact state has no entropy, and the arrow is a 0/0. Coarse-graining removes the singularity: averaging over the bin turns hidden information into heat, the Ch.65 fee, paid by the lens to itself. Reversible law + coarse lens = the Second Law, measured. Reversible law + fine lens = the identity, measured. The two results are the same words in two lenses.

6. Connections to Prior 0/0 Singularities

Arrow of time (Ch.48): the lens, not the law, is the clock. Eternal return (Ch.57): Zermelo's recurrence is defused by the lens. Reversible cycle (Ch.65): Delta S = 0 needs a fine eye. Noether (Ch.66): the same reversal test - the law returns exactly. Meaning (Ch.61): information is not lost, only refracted.

References

[1] Boltzmann (1872), Weiter Studien uber das Warmegleichgewicht
[2] Loschmidt (1876), reversal objection (reversibility paradox)
[3] Zermelo (1896), recurrence objection to the H-theorem
[4] Gibbs (1902), Elementary Principles in Statistical Mechanics
[5] Arnold & Avez (1968), ergodic problems of classical mechanics
[6] Puno, "The Removable Singularity" (2026), chapters 48-66